

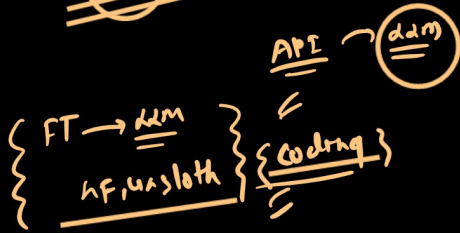
Decoder

- ① Positional encoding
- ② Self Attention $\longrightarrow$ Multihead Self Attention
- ③ layer normalization

$\approx$ { Cross Attention
 $\succ$ { Self mask attention

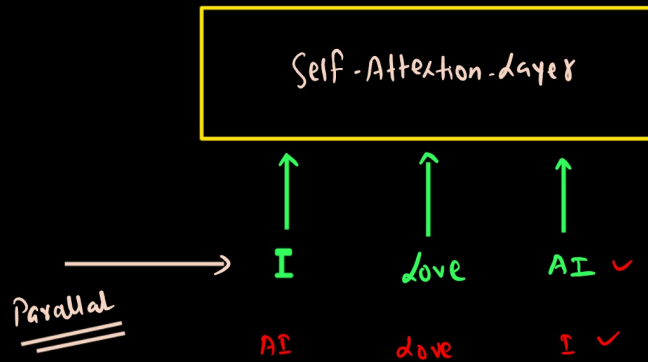
end2end flow encoder $\longrightarrow$ decoder

2-3



Positional encoding

Problem statement → Data → Parallelly Soln → Sequence → PE



Positional Encoding (Definition)

Positional encoding is a technique used to inject information about the position (order) of tokens into their embeddings, so that the model can understand sequence order.

Why it is needed?

Transformers process all tokens in parallel

So they don't know order by default

Example:

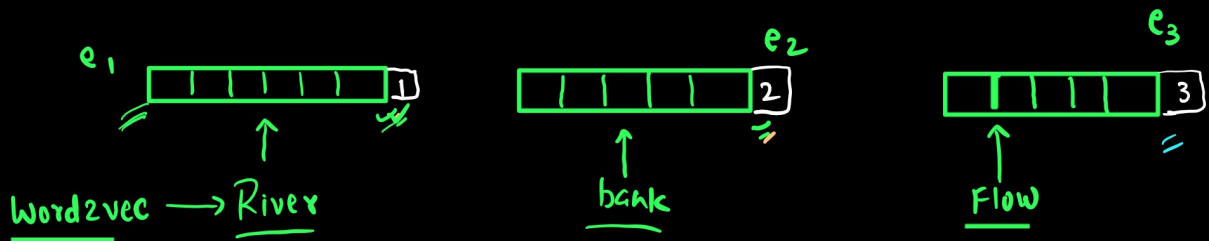
- "I love AI"
- "AI love I"

Without positional encoding → both look same to model

Dog bites man

man bites dog

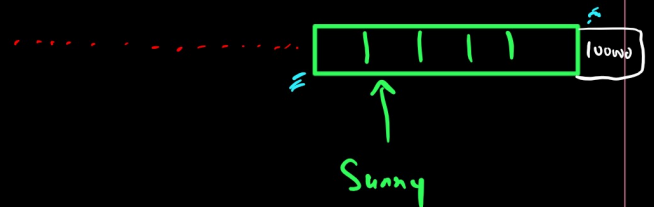
Self-Attention-Layer



Real time data

Self
100000

No-limit $\leq$



Simple Approach

Self $\rightarrow$ 4 token

I Love AI Much
1 2 3 4

Self $\rightarrow$ 4w token

I Love AI Much again You
1 2 3 4 5 4w

Self $\rightarrow$ Limitless

n token

1L, 10L, ...

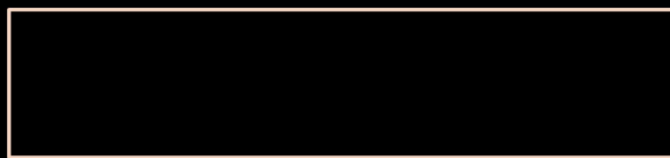
Drawback

① No of token could be limitless

② Discrete (1, 2, 3, 4, 5, ..., 1000, 10000) $\rightarrow$ vector

NN

(training might become unstable)



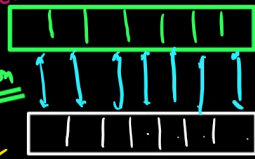
← this vector having a knowledge of pos

Positional encoding = PE1

PE vector

Addition

e_1

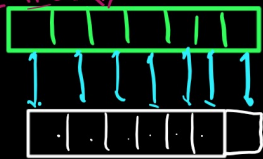


great

Word 2 vec

PE2

e_2



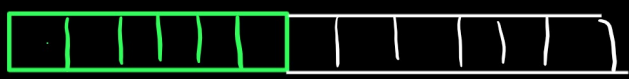
indig

Same Dim of vector

Concatination

(APPEND)

Dim ↑↑↑↑↑



10-Dim

10-Dim

NN / Attention

more computational Power

ADDITION

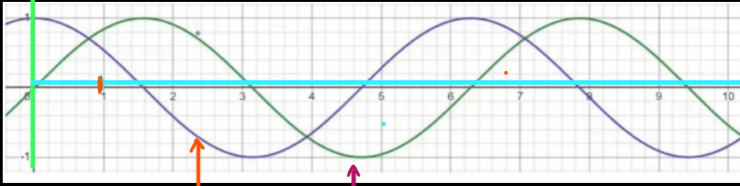
$e_1 + PE1$



Same dim of vector

PE1

Encoded value



cos
[-1 to +1]

sin $\rightarrow$ [-1 to +1]

PE2

9-10th class

School

Pythagoras

TRIGONOMETRY
Sin, cos, tan

Position
of the word



Sin | cos

original RP
512

))))
= word2vec

9vec
1

radia
2

~~Descarte~~

Continue

E1



sin(Pos)
sin(1) $\rightarrow$ 0.84
cos(1) $\rightarrow$ 0.54
cos(Pos)
 $\approx$ [0.84 | 0.54]
2

E2



sin(Pos)
sin(2) $\rightarrow$ 0.9
cos(2) $\rightarrow$ -0.4
cos(Pos)
[0.9 | -0.4] $\approx$
2

PE2

$$PE_{(pos, 2i)} = \sin(pos / 10000^{2i/d_{model}}) \quad \checkmark \quad \sin(Pos) \longrightarrow \sin(Pos \mid 10000^{2i/d_{model}})$$

$$PE_{(pos, 2i+1)} = \cos(pos / 10000^{2i/d_{model}}) \quad \checkmark \quad \cos(Pos) \longrightarrow \cos(Pos \mid 10000^{2i/d_{model}})$$

Pos $\longrightarrow$ Position

d_{model} $\longrightarrow$ Dim of embedding model

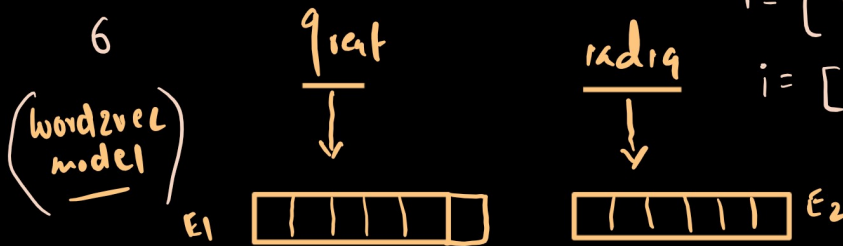
i $\longrightarrow$ Range

$$i = [\text{Low Range} - \text{Upper Range}]$$

$$= [0 - (\frac{d_{model}}{2} - 1)]$$

$$i = [0 - (\frac{6}{2} - 1)]$$

$$i = [0 - 2] \longrightarrow \checkmark 0, 1, 2 \}$$



4 tokens

I Love my country
 Pos $\longrightarrow$ 0 1 2 3

Dim → 6

Pos = 0
great

E₁



E₂



Pos = 1
india

wordvec

d_{model} → 6 =

$$i = \left[0 - (6/2 - 1) \right]$$

$$= [0 - 2]$$

$$i = 0, 1, 2$$

sin/cos cos/sin sin/cos → 6

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{model}})$$

$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{model}})$$

PE₂

For i = 0 → Pos → 0

$$PE(0, 0) = \sin(0/10000^0) = 0$$

$$PE(0, 1) = \cos(0/10000^0) = 1$$

$$PE(1, 0) = \sin(1/10000^0) = 0.84$$

$$PE(1, 1) = \cos(1/10000^0) = 0.54$$

For i = 1 2x1

$$PE(0, 2) = \sin(0/10000^{1/3}) = 0$$

$$PE(0, 3) = \cos(0/10000^{1/3}) = 1$$

For i = 1

$$PE(1, 2) = \sin(1/10000^{1/3}) = 0.04$$

$$PE(1, 3) = \cos(1/10000^{1/3}) = 0.99$$

For i = 2

$$PE(0, 4) = \sin(0/10000^{2/3}) = 0$$

$$PE(0, 5) = \cos(0/10000^{2/3}) = 1$$

For i = 2

$$PE(1, 4) = \sin(1/10000^{2/3}) = 0.01$$

$$PE(1, 5) = \cos(1/10000^{2/3}) = 0.98$$

PE₁ [0 | 1 | 0 | 1 | 0 | 1]

PE₂ → [0.84 | 0.54 | 0.04 | 0.99 | 0.01 | 0.98]

Dim 6



x₁



x₂

→ TODAY 1 AM TEACHING PE

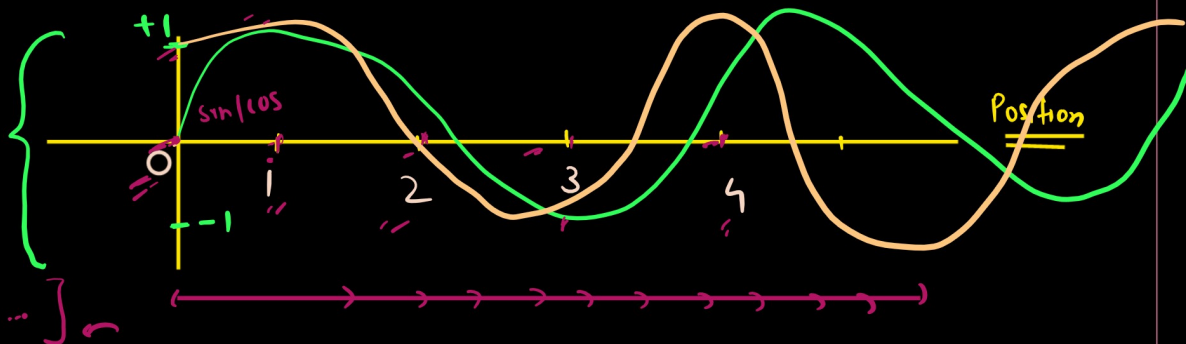
tokens → 5

Pos → 0 1 2 3 4

~~discrete~~
continuous (sin/cos)

encoded value

seq info



② Self Attention → Multihed Attention

Money bank grow

Similarity → Dot Product

$$\text{bank} = w_1 * \text{Money} + w_2 * \text{bank} + w_3 * \text{grow}$$

(q, k)

$$\text{bank} = \underbrace{\left[\underbrace{e_{\text{bank}}}_{q} \cdot \underbrace{e_{\text{money}}}_{k} \right] \text{Money}}_{s_1} + \underbrace{\left[\underbrace{e_{\text{bank}}}_{q} \cdot \underbrace{e_{\text{bank}}}_{k} \right] * \text{bank}}_{s_2} + \underbrace{\left[\underbrace{e_{\text{bank}}}_{q} \cdot \underbrace{e_{\text{grow}}}_{k} \right] * \text{grow}}_{s_3}$$

SOFTMAX

$$w_1 = \frac{e^{s_1}}{e^{s_1} + e^{s_2} + e^{s_3}}$$

$$w_2 = \frac{e^{s_2}}{e^{s_1} + e^{s_2} + e^{s_3}}$$

$$w_3 = \frac{e^{s_3}}{e^{s_1} + e^{s_2} + e^{s_3}}$$

value vector (v)

$$w_1 * \text{money}$$

(v)

$$w_2 * \text{bank}$$

(v)

$$w_3 * \text{grow}$$

{Self → (Q, K, V)}

bank

Contextual vector

$$\begin{bmatrix} 1 & 2 \end{bmatrix} \quad \begin{bmatrix} 3 & 4 \end{bmatrix}$$

$$\begin{matrix} \times \\ \begin{bmatrix} 1 & 2 \end{bmatrix} \\ a_1 \quad b_1 \end{matrix} \quad \begin{matrix} \times \\ \begin{bmatrix} 3 \\ 4 \end{bmatrix} \\ a_2 \quad b_2 \end{matrix}$$

$$(1 \times 3) + (2 \times 4) = 11 \quad \text{Scalar value}$$

$a_1 \times a_2 \quad b_1 \times b_2$

Self-Attention $\rightarrow$ Q, K, V

$$D = \{a:1, b:2, c:3\}$$

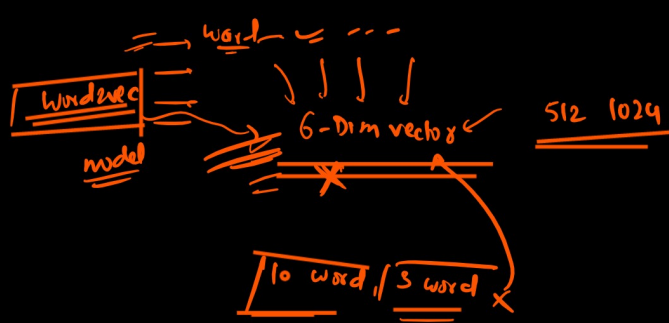
Analogy

$$= D[a] \rightarrow \text{o/p} = 1$$

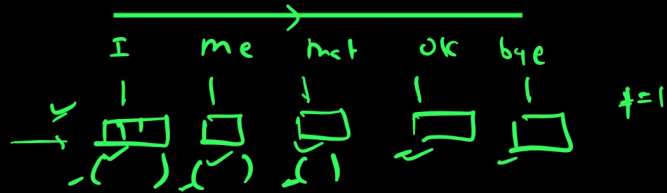
query key value

Self-Attention $\rightarrow$ extend

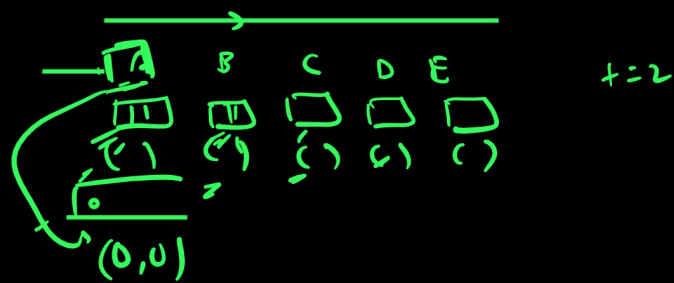
- in self-attention
- Cross Attention
- masked " " "



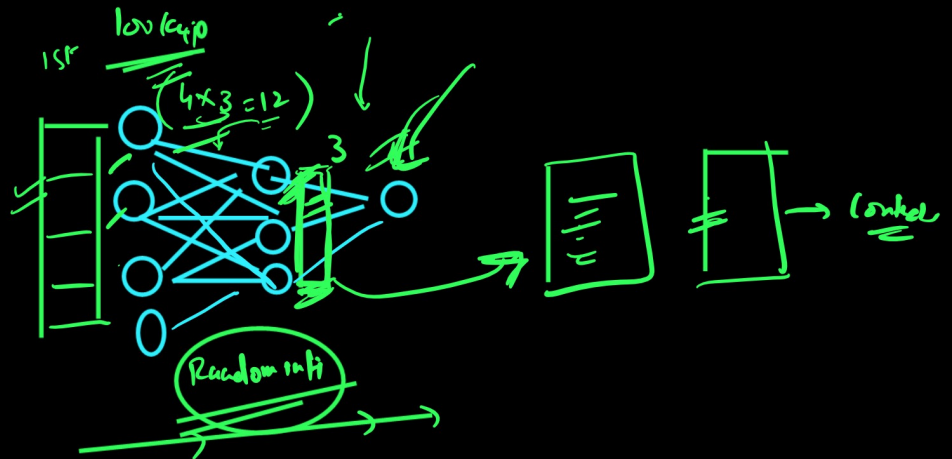
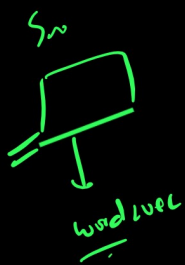
Sent → 5 word

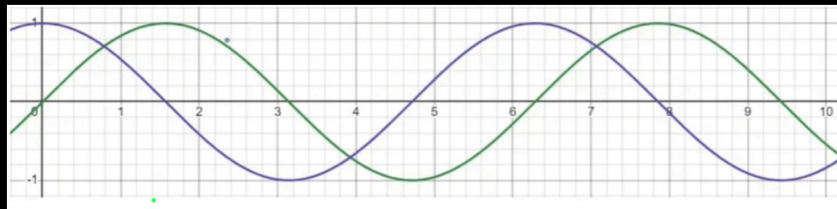


S2 → 10 word →



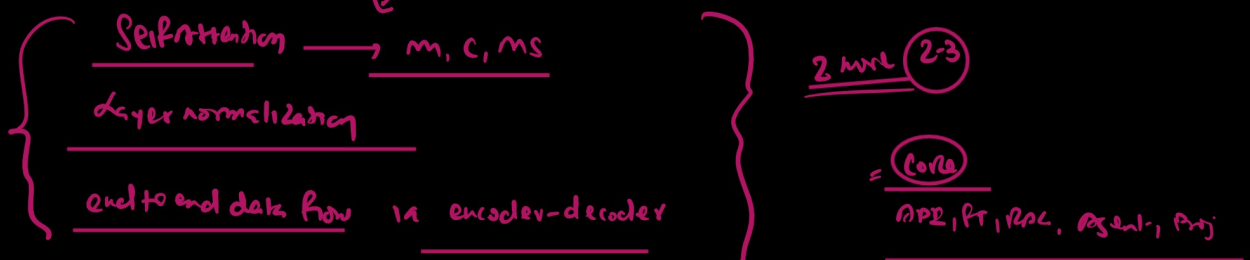
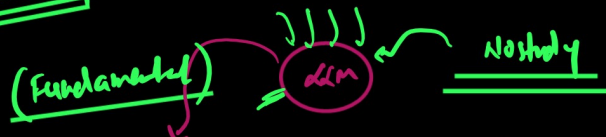
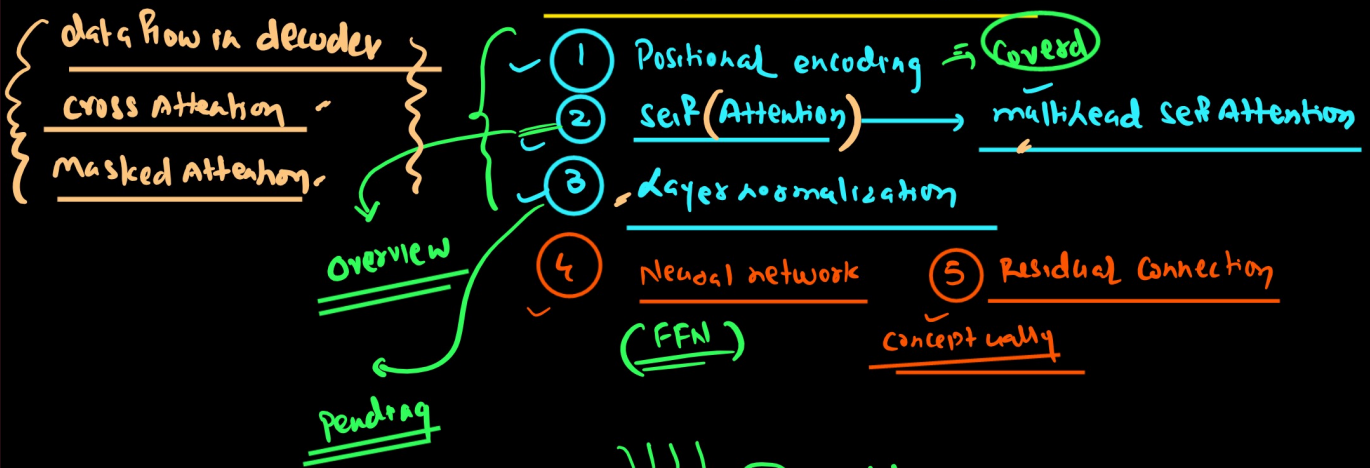
Google → W13





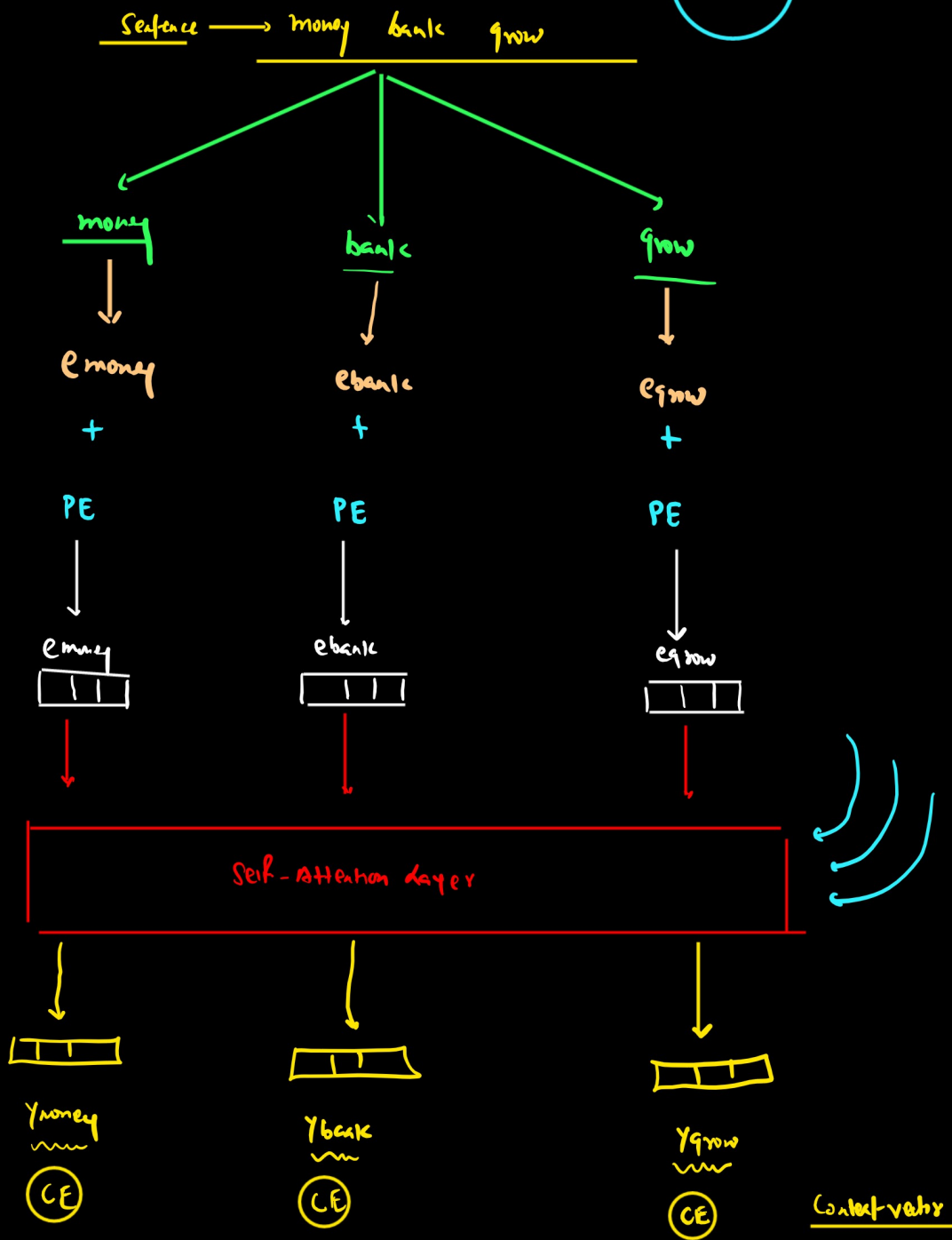
- ① Encoder
- ② Decoder

How the data flows in encoder



Self Attention

Parallel



Similarity score

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Self-Attention → Contextual vector

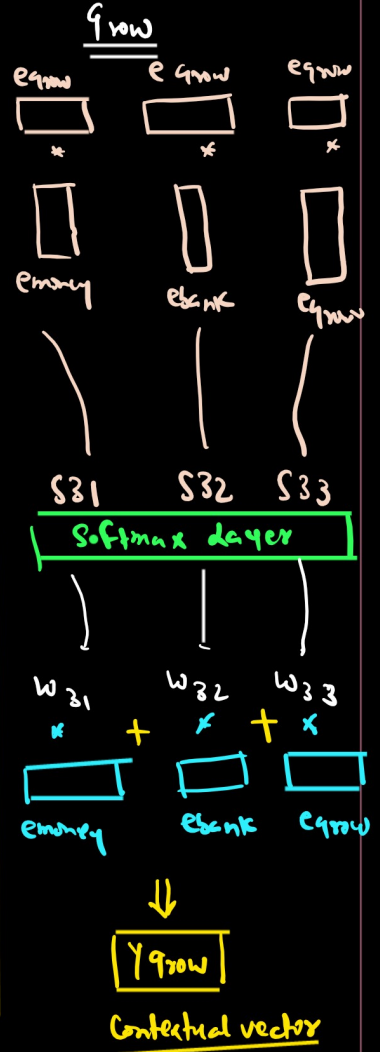
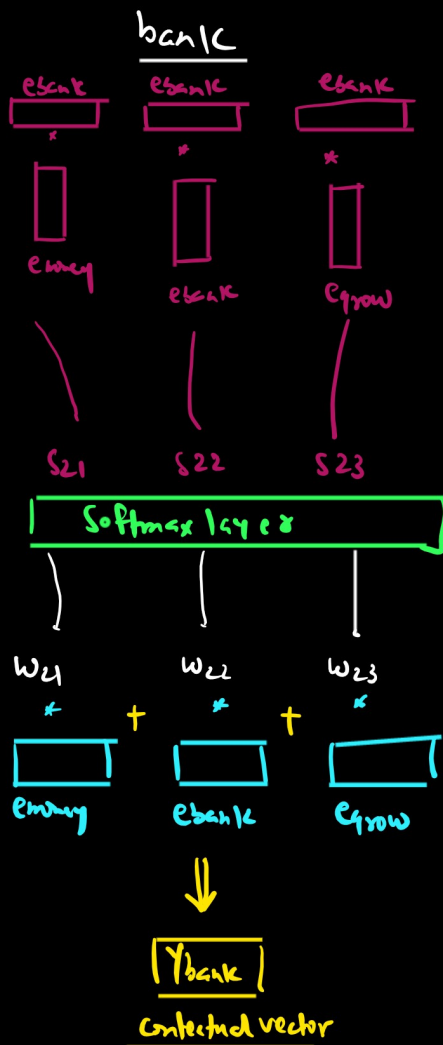
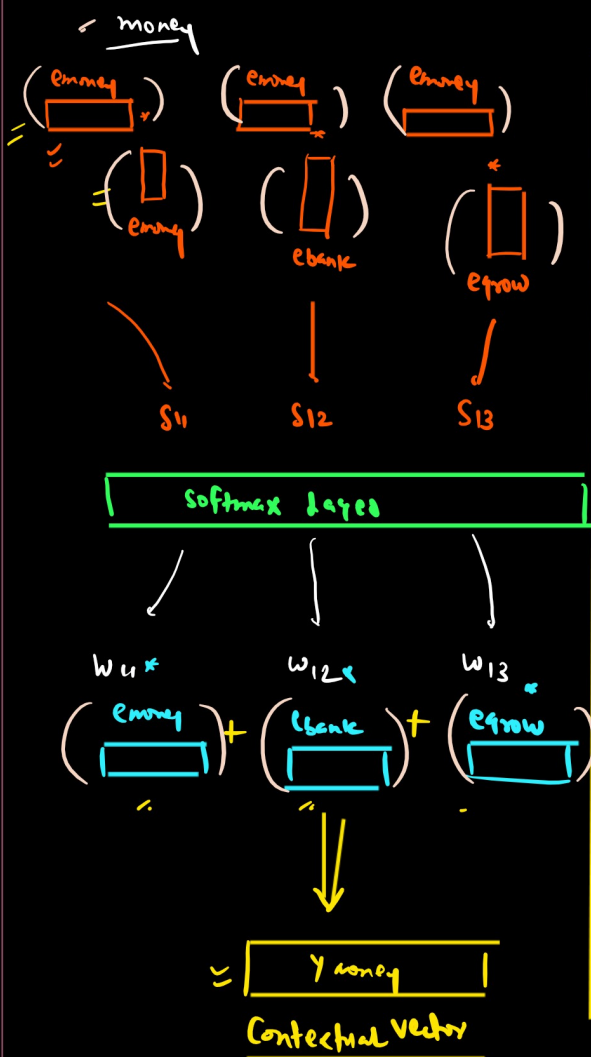
money bank grow $\longrightarrow$ How this text is being processed using Self Attention

Find out the Relation of each word with word

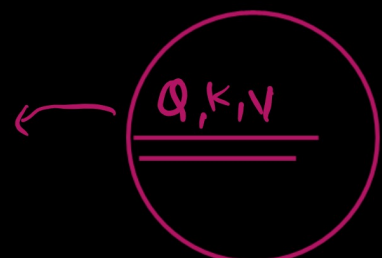
$$\begin{aligned}
 \text{money CE} &= \overbrace{w_{11}}^{\text{similarity score}} * e_{\text{money}} + w_{12} * e_{\text{bank}} + w_{13} * e_{\text{grow}} \\
 \text{bank CE} &= w_{21} * e_{\text{money}} + w_{22} * e_{\text{bank}} + w_{23} * e_{\text{grow}} \\
 \text{grow CE} &= w_{31} * e_{\text{money}} + w_{32} * e_{\text{bank}} + w_{33} * e_{\text{grow}}
 \end{aligned}$$

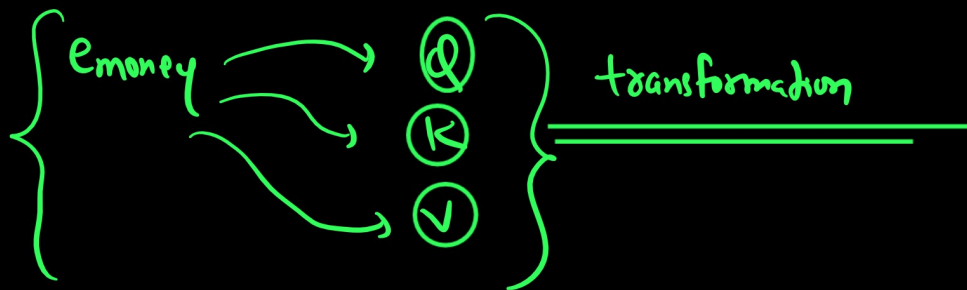
Dot Product
 $\uparrow$

(SS)



Same vector $\longrightarrow$ (multiple times)
(this is confusing)

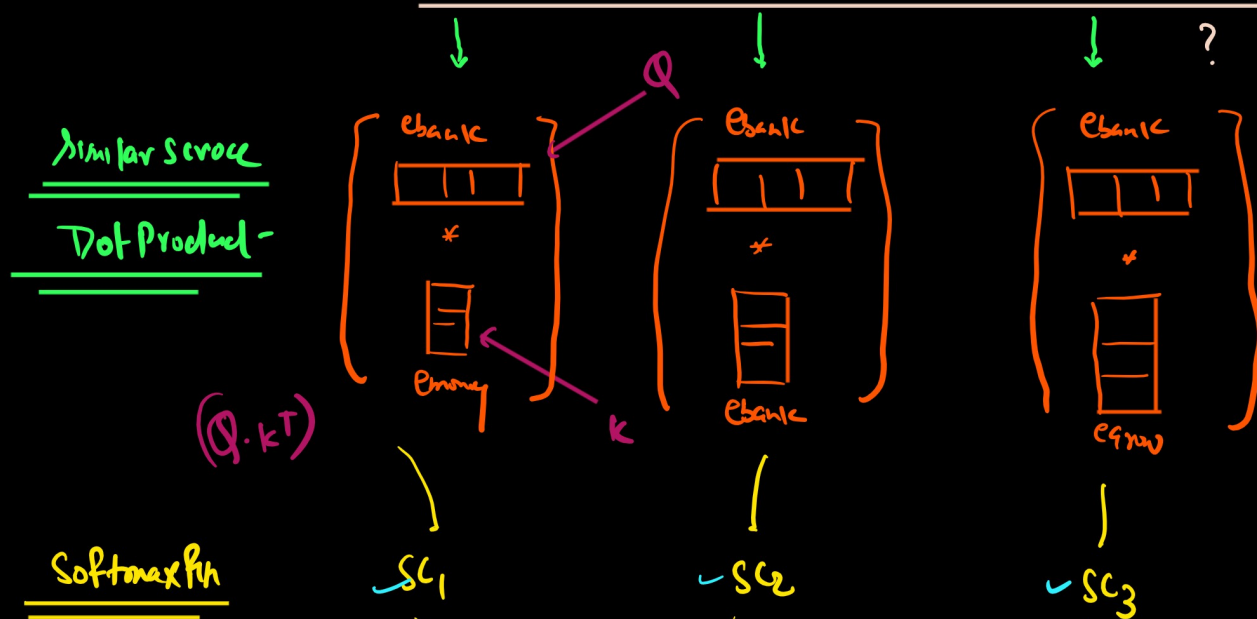




money | bank | grow $\xrightarrow{CE}$ self-Attention?

map this word in context of other word $\Leftarrow$

$$bank_{CE} = w_{11} * emoney + w_{12} * e_{bank} + w_{13} * e_{grow}$$



Softmax layer

$$w_1 = \frac{e^{sc_1}}{e^{sc_1} + e^{sc_2} + e^{sc_3}}$$

$$w_2 = \frac{e^{sc_2}}{e^{sc_1} + e^{sc_2} + e^{sc_3}}$$

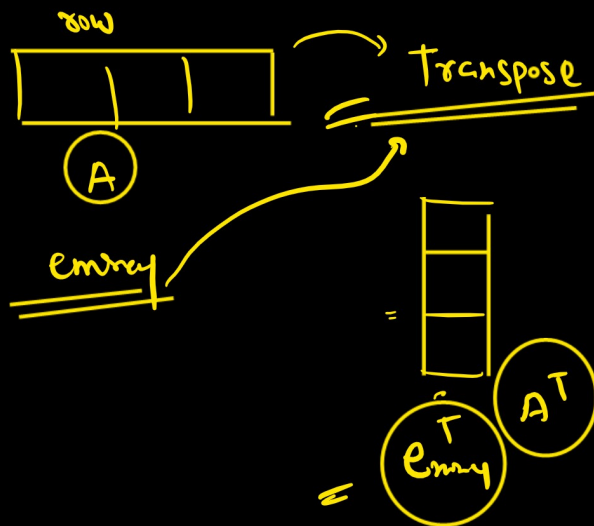
$$w_3 = \frac{e^{sc_3}}{e^{sc_1} + e^{sc_2} + e^{sc_3}}$$

$$w_1 * \begin{bmatrix} | & | & | \end{bmatrix}_{emoney} + w_2 * \begin{bmatrix} | & | & | \end{bmatrix}_{ebank} + w_3 * \begin{bmatrix} | & | & | \end{bmatrix}_{egrow}$$

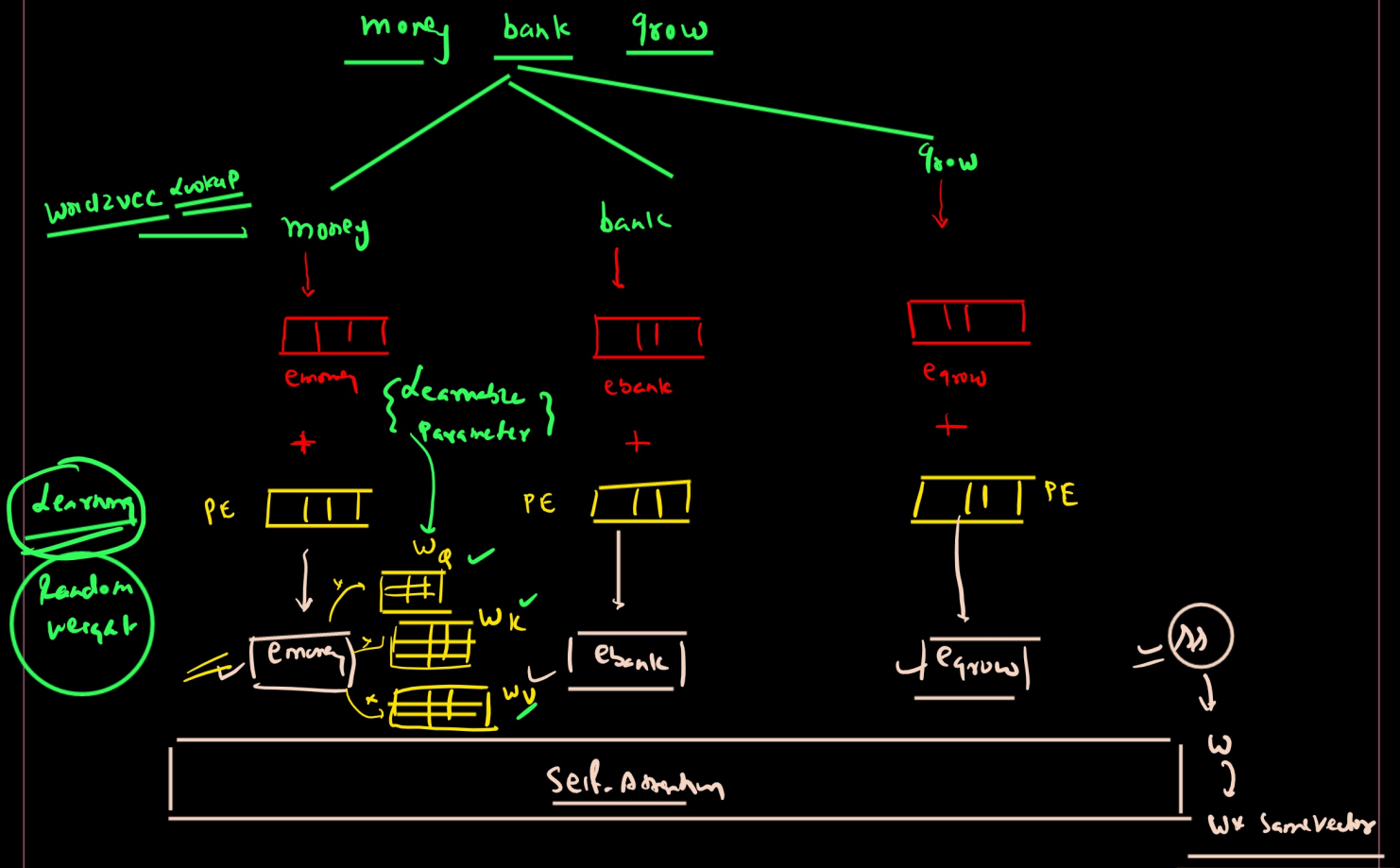
$$= \text{Softmax} \left(\frac{Q \cdot K^T}{\sqrt{d_k}} \right) * \underline{\underline{V_{key}}}$$

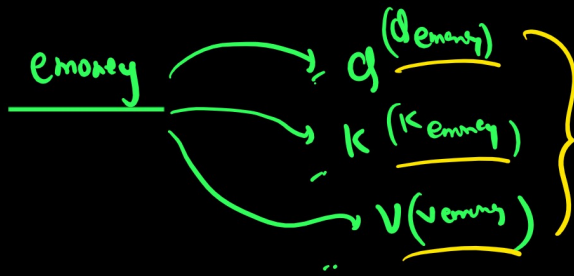
$\begin{bmatrix} | & | & | \end{bmatrix}_{ebank}$

$\leftarrow$ Contextual vector

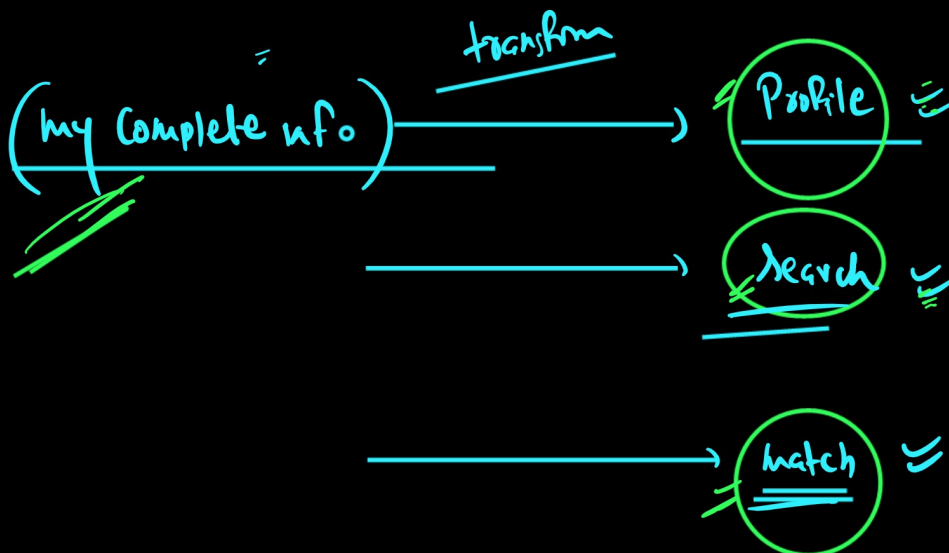
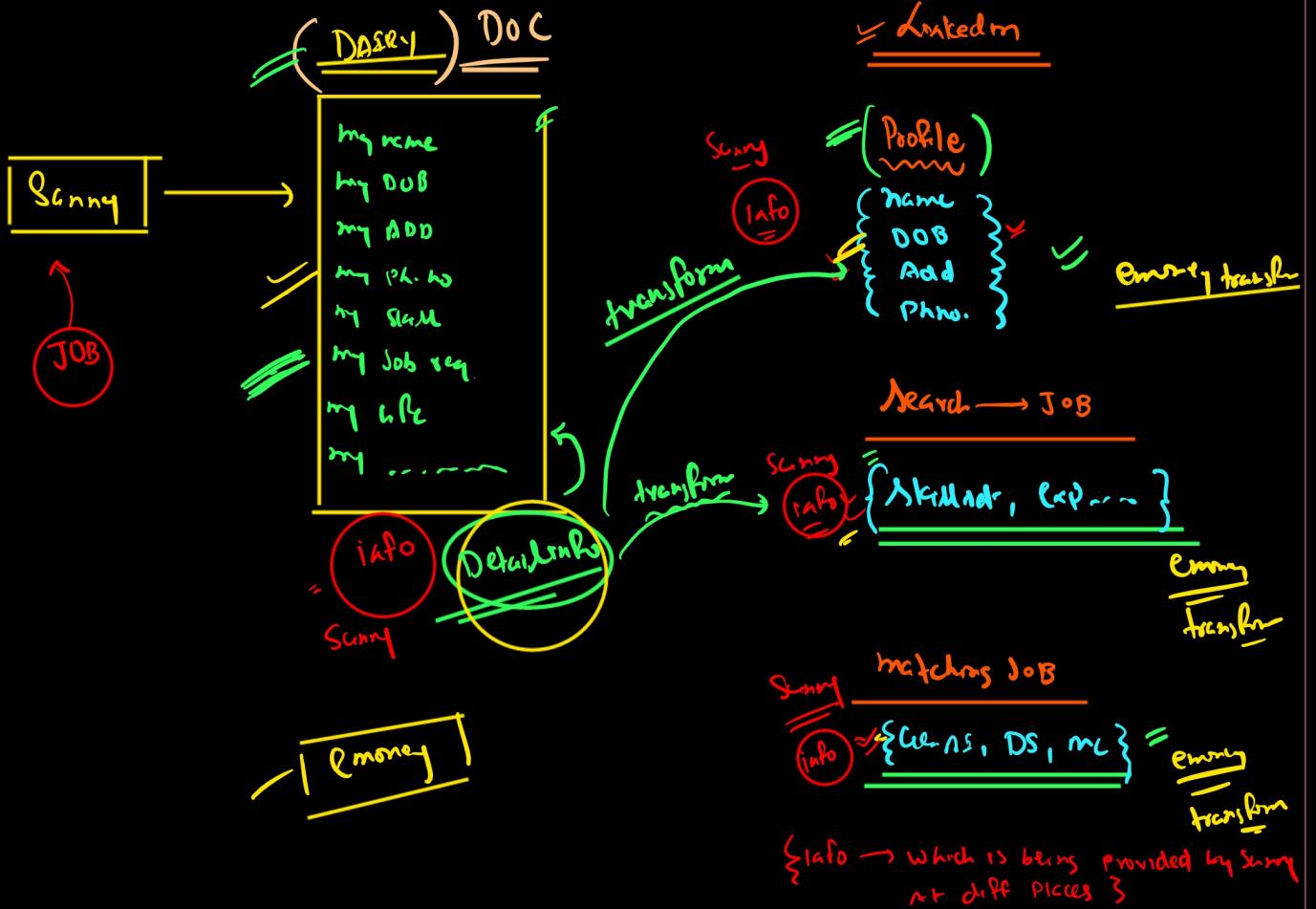


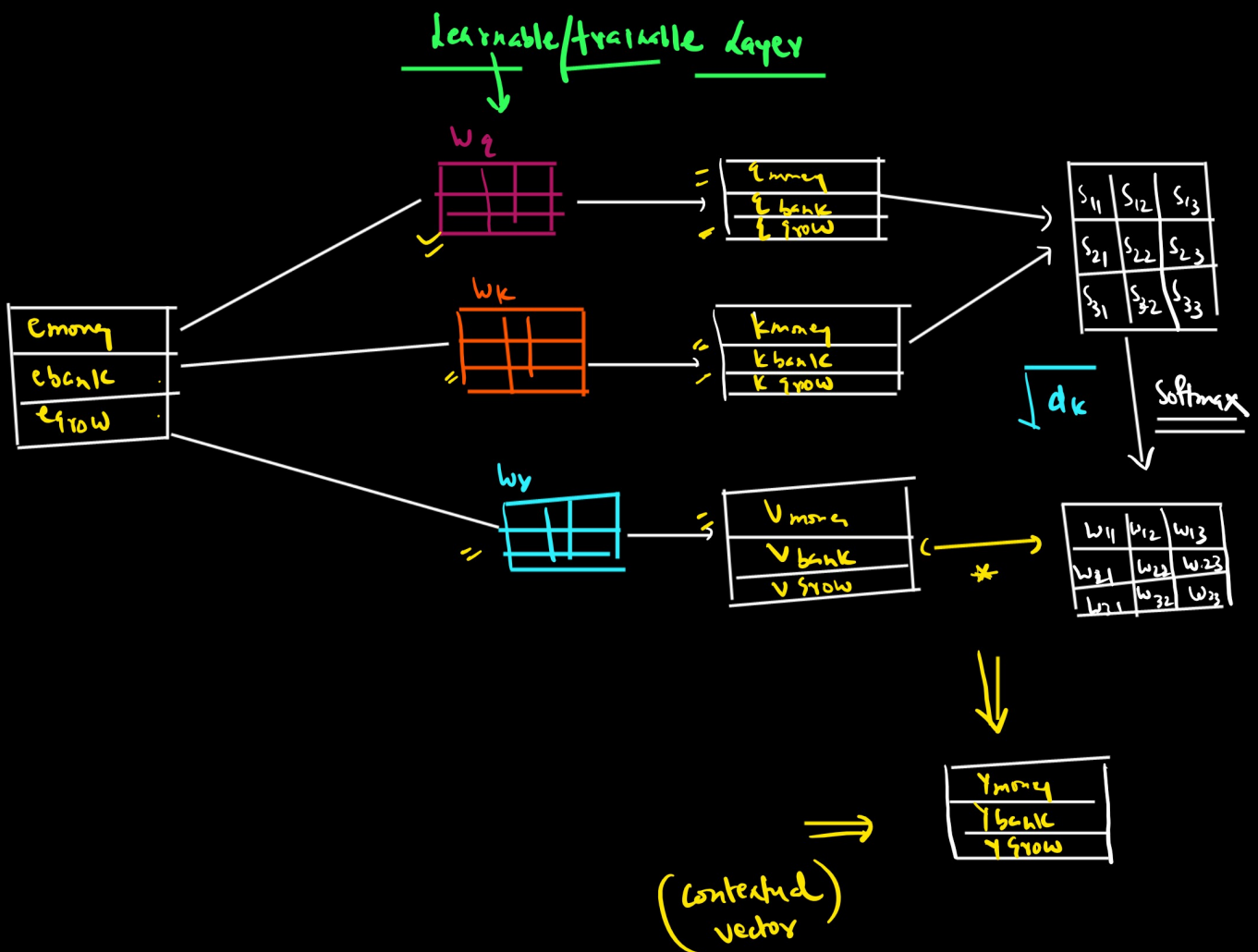
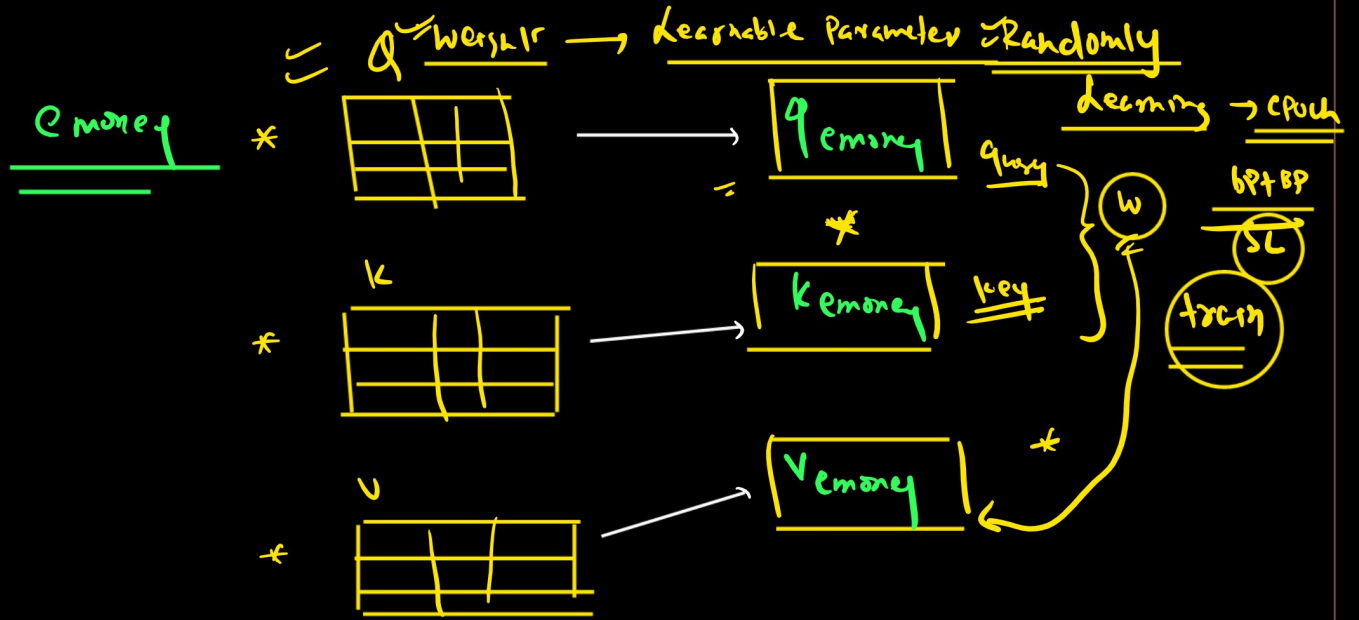
Dot Product
horizontal * vertical

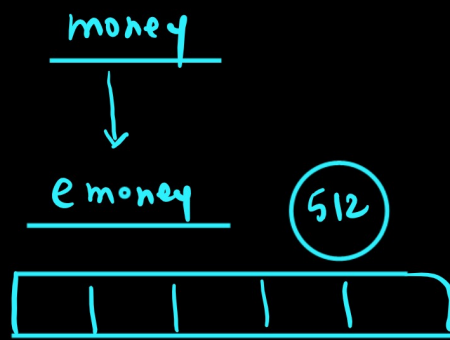




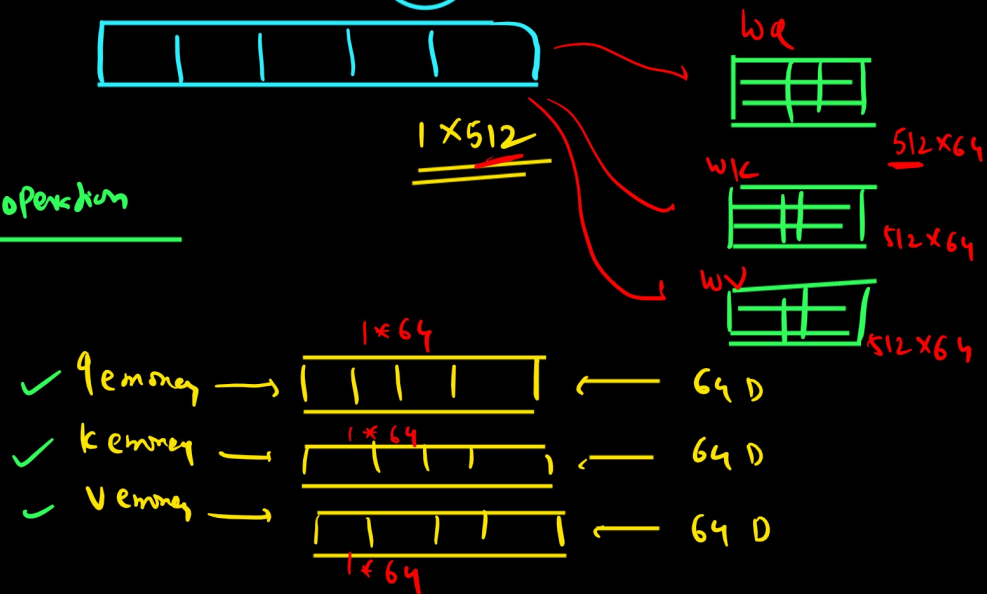
Analogy







basic matrix operation



(64)

q_{e money} * k_{e money} = value ?

↓
✗

Softmax

$\left(\frac{\text{value}}{\sqrt{d_k}} \right) = (\text{value}')$

↓ $\sqrt{64} = 8$

dimension

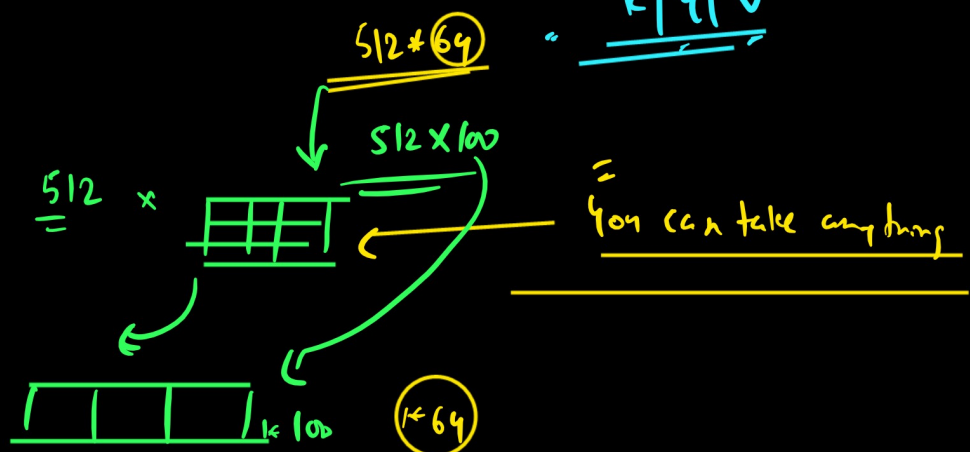
$\left(\frac{\text{normalized}}{\text{value}} \right)$

k | q | v

modified Self dimension reduction

cross / track

tomorrow



$$q \times w_q$$

$$k \times w_k$$

$$v \times w_v$$

= Final summary find doc